

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“Jnana Sangama”, Machhe, Belagavi, Karnataka-590018



Lab Experiment Record

Project Management with Git [BCSL358C]

Submitted in partial fulfillment towards AEC of 3rd semester of

Bachelor of Engineering
in

Computer Science and Engineering
(Artificial Intelligence & Machine Learning)

Submitted by

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GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN

(Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi & Govt. of Karnataka)

K.R.S ROAD, METAGALLI, MYSURU-570016, KARNATAKA
(Accredited by NAAC)

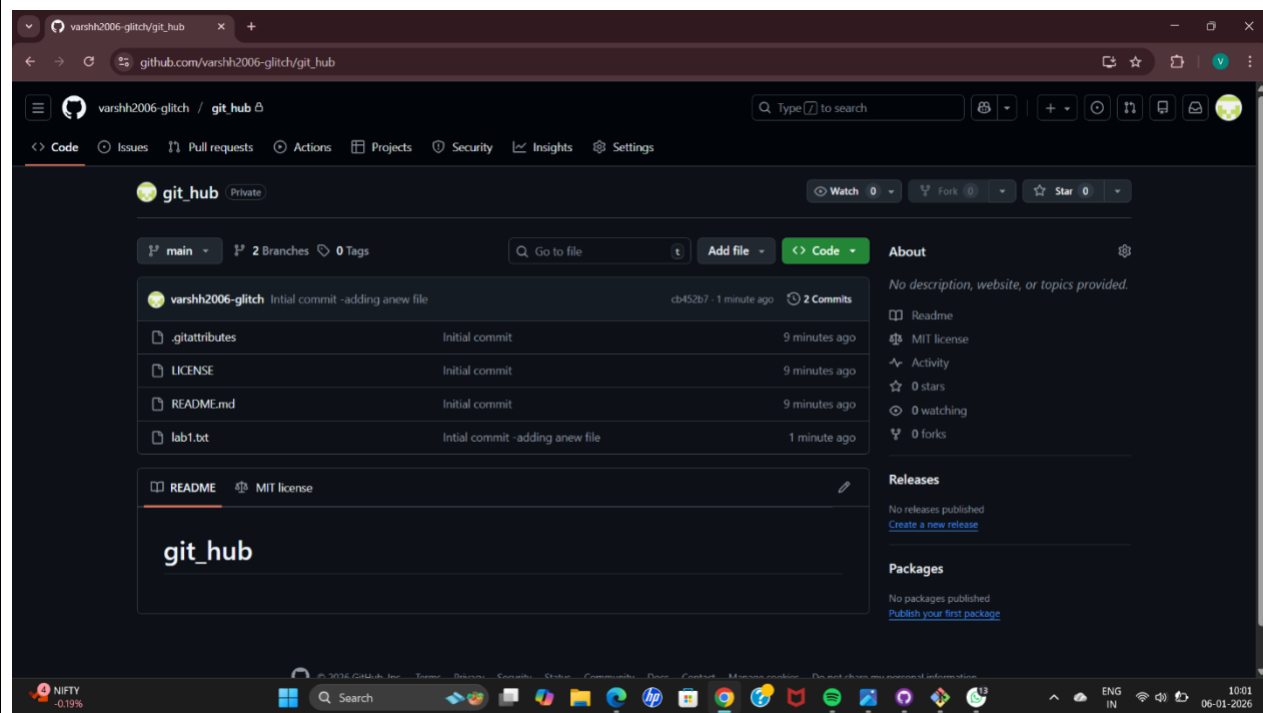
EXPERIMENT 1: SETTING UP AND BASIC COMMANDS

PROCEDURE

1. Open Git Bash and move to the project folder created on the Desktop using the cd command.
2. Initialize the folder as a Git repository using the git init command.
3. Create a new text file using Notepad to store the experiment content.
4. Add the created file to the staging area using the git add command.
5. Check the status of the repository using the git status command.
6. Commit the staged file to the local repository using the git commit command.
7. Link the local repository to the remote GitHub repository using the git remote add origin command.
8. Rename the default branch to main using the git branch M main command.
9. Synchronize the local repository with the remote repository using the git pull --rebase command.
10. Push the committed files from the local repository to the GitHub repository using the git push command.

GIT REPO URL :

https://github.com/varshh2006-glitch/git_hub.git



MINGW64:/c/Users/Varshitha S/git_hub

```
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~ (master)
$ git clone https://github.com/varshh2006-glitch/git_hub.git
Cloning into 'git_hub'...
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 0), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (5/5), done.

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~ (master)
$ cd git_hub

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$ touch lab1.txt

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$ git add lab1.txt

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$ git commit -m"Intial commit -adding anew file"
[main cb452b7] Intial commit -adding anew file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 lab1.txt

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 296 bytes | 296.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/varshh2006-glitch/git_hub.git
c636f36..cb452b7  main -> main

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$
```

COMMANDS USED:

git init notepad
lab1.txt git add

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```
lab1.txt git
status
git commit -m "Experiment 1: Initial repository setup and first commit" git
remote add origin https://github.com/chinmayimohan2006-web/github.git git
branch -M main git pull --rebase origin main
git push -u origin main
```

EXPERIMENT 2: CREATING AND MERGING BRANCHES

PROCEDURE

1. Open Git Bash and move to the project directory.
2. Switch to the main branch using the git checkout command.
3. Create a new branch named feature-branch and switch to it.
4. Modify an existing file using Notepad.
5. Add the modified file to the staging area.
6. Commit the changes in the feature branch.
7. Switch back to the main branch.
8. Merge the feature-branch into the main branch.

COMMANDS USED:

```
git checkout main git checkout -b feature-branch notepad
lab1.txt git add lab1.txt git commit -m "Experiment 2:
Changes in feature branch" git checkout main git merge
feature-branch
```

```
c636f36..cb452b7 main -> main
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (main)
$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git add .

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git commit -m "Merge the feature-branch into master"
On branch feature-branch
nothing to commit, working tree clean

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git checkout master
branch 'master' set up to track 'origin/master'.
Switched to a new branch 'master'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git merge feature-branch
Updating c636f36..cb452b7
Fast-forward
 lab1.txt | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 lab1.txt

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$
```



EXPERIMENT 3: CREATING AND MERGING BRANCHES

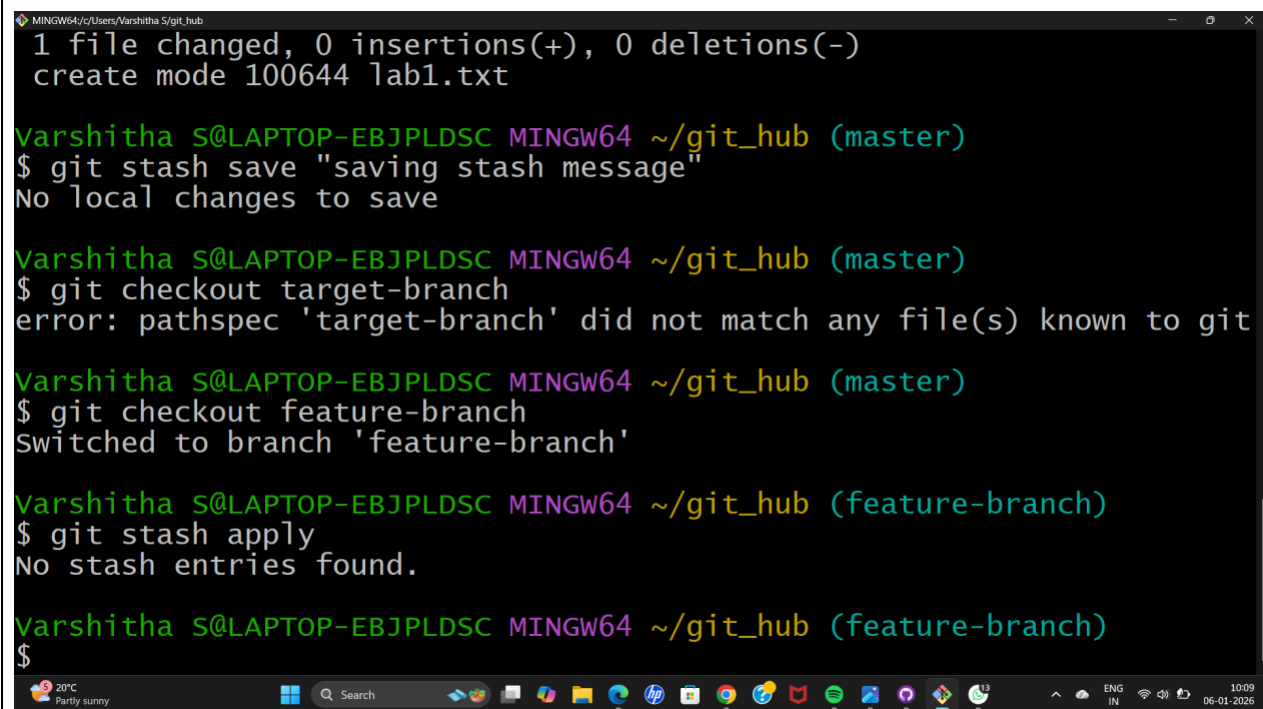
PROCEDURE:

1. Open Git Bash and move to the project directory.
2. Modify an existing file using Notepad.
3. Check the repository status using the git status command.
4. Save the uncommitted changes using the git stash command.
5. Verify that the working directory is clean.
6. Switch to another branch.

7. Apply the stashed changes using the git stash apply command.
8. Check the repository status again.

COMMANDS USED:

notepad lab1.txt , git status , git stash , git checkout feature1 , git stash apply , git status



```
MINGW64/c:/Users/Varshitha S/git_hub
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 lab1.txt

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git stash save "saving stash message"
No local changes to save

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git checkout target-branch
error: pathspec 'target-branch' did not match any file(s) known to git

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git checkout feature-branch
Switched to branch 'feature-branch'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git stash apply
No stash entries found.

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$
```

EXPERIMENT 4: COLLABORATIONS AND REPORT REPOSITORIES

PROCEDURE:

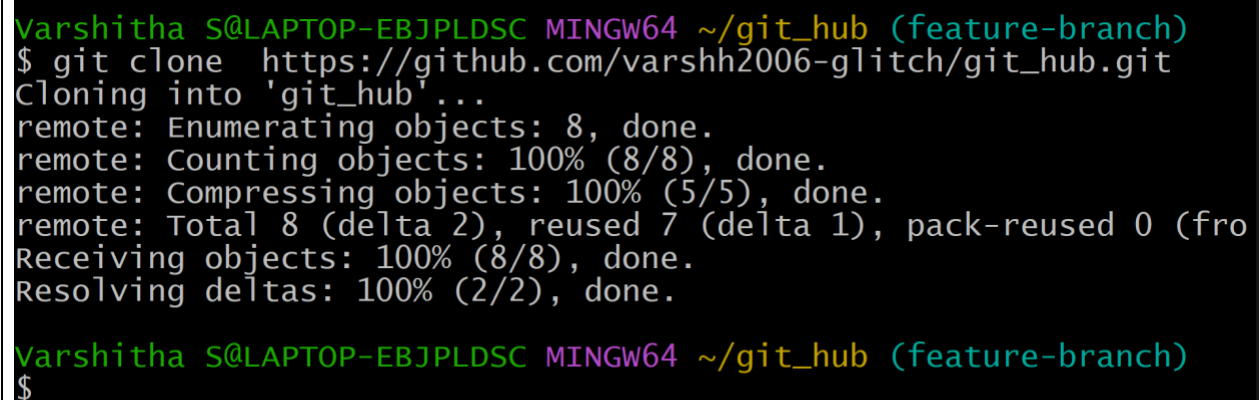
1. The terminal was navigated to the Desktop using the cd command.

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2. The remote GitHub repository was copied to the local system using the git clone command.
3. The cloned repository folder was accessed using the cd command.
4. The files present in the repository were displayed using the ls command.
5. The current status of the repository was checked using the git status command.

COMMANDS USED:

```
cd ~/OneDrive/Desktop git
clone <repository-URL>
cd <repository-folder>
ls git
status
```



```
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git clone https://github.com/varshh2006-glitch/git_hub.git
Cloning into 'git_hub'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 8 (delta 2), reused 7 (delta 1), pack-reused 0 (fro
Receiving objects: 100% (8/8), done.
Resolving deltas: 100% (2/2), done.

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$
```

EXPERIMENT 5: COLLABORATIONS AND REPORT REPOSITORIES

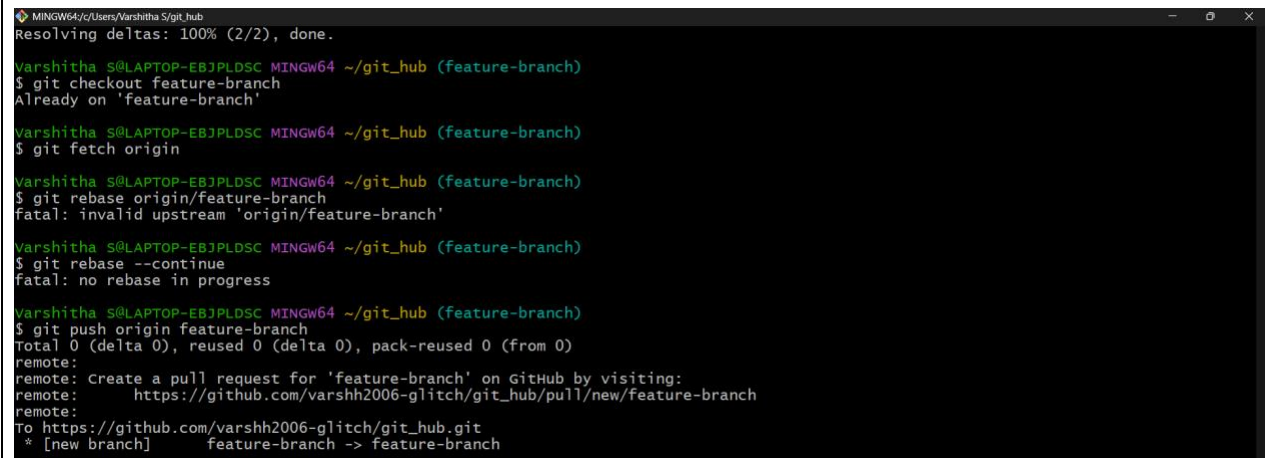
PROCEDURE:

1. The terminal was navigated to the local GitHub repository directory.
2. The current branch and working tree status were verified using the git status command.
3. The latest changes from the remote repository were fetched using the git fetch origin command.
4. The local branch was synchronized with the remote main branch using the git rebase origin/main command.

5. The repository status was checked again to ensure that the working tree was clean and up to date.

COMMANDS USED:

cd ~/OneDrive/Desktop/github , git status , git status , git fetch origin, git rebase origin/main



```

MINGW64~/c/Users/Varshitha S/git_hub
Resolving deltas: 100% (2/2), done.

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git checkout feature-branch
Already on 'feature-branch'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git fetch origin

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git rebase origin/feature-branch
fatal: invalid upstream 'origin/feature-branch'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git rebase --continue
fatal: no rebase in progress

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git push origin feature-branch
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: create a pull request for 'feature-branch' on GitHub by visiting:
remote:   https://github.com/varshh2006-glitch/git_hub/pull/new/feature-branch
remote:
To https://github.com/varshh2006-glitch/git_hub.git
 * [new branch]      feature-branch -> feature-branch
  
```

EXPERIMENT 6: COLLABORATION AND REMOTE REPOSITORIES

PROCEDURE:

1. The list of available branches was displayed using the git branch command.
2. The current working branch was switched to the main branch using git checkout main.
3. The changes from the feature branch were merged into the main branch using the git merge feature1 command.
4. The status of the repository was verified using the git status command to ensure successful merge.

COMMANDS USED:

git branch, git checkout main, git , git merge feature1, git status

```
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (feature-branch)
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git merge feature-branch -m"feature-branch into master"
Already up to date.

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$
```

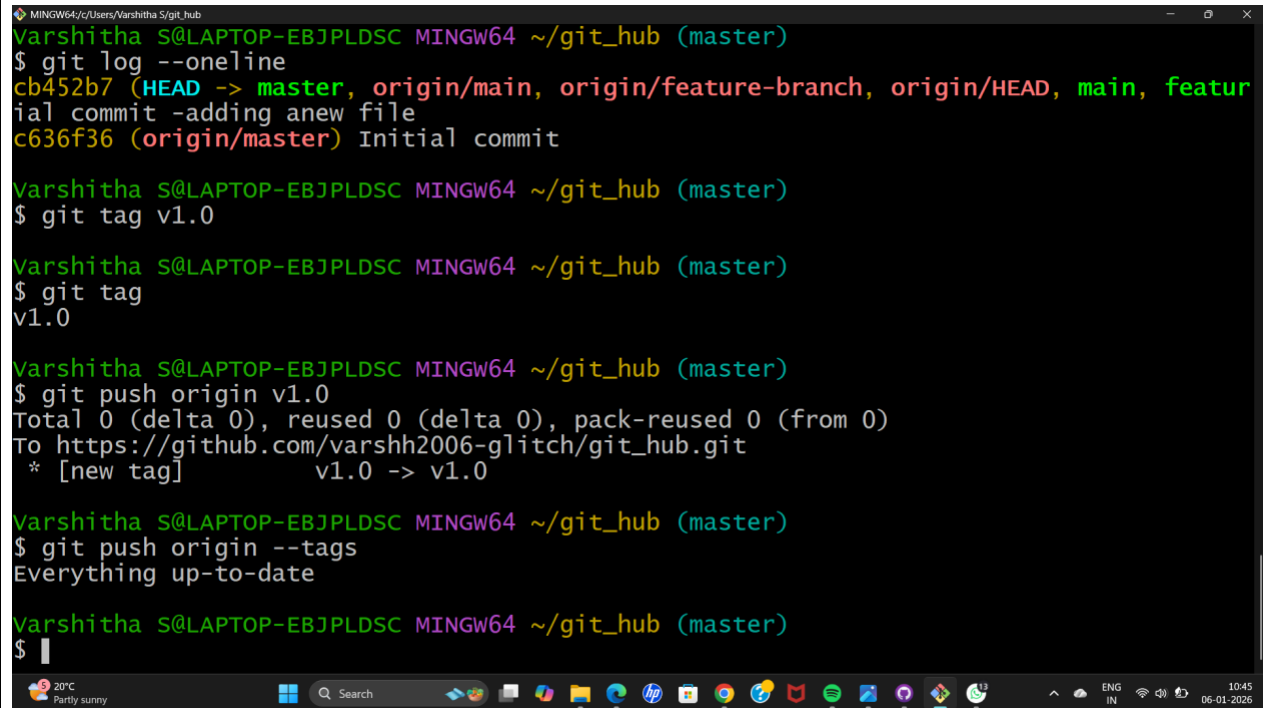
EXPERIMENT 7: GIT TAGS AND RELEASES

PROCEDURE:

1. The working branch was verified and switched to the main branch using the git checkout main command.
2. The current repository status was checked using git status.
3. A tag named v1.0 was created using the git tag command to mark an important version of the project.
4. The list of available tags was displayed using the git tag command.
5. The commit history was viewed using git log --oneline to confirm that the tag was attached to the correct commit.

COMMANDS USED:

git checkout main , git status, git tag v1.0, git tag , git log --oneline



```
MINGW64/c/Users/Varshitha S/git_hub
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git log --oneline
cb452b7 (HEAD -> master, origin/main, origin/feature-branch, origin/HEAD, main, feature1)
ial commit -adding anew file
c636f36 (origin/master) Initial commit

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git tag v1.0

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git tag
v1.0

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git push origin v1.0
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/varshh2006-glitch/git_hub.git
 * [new tag]          v1.0 -> v1.0

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git push origin --tags
Everything up-to-date

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$
```

EXPERIMENT 8: ADVANCED GIT OPERATIONS

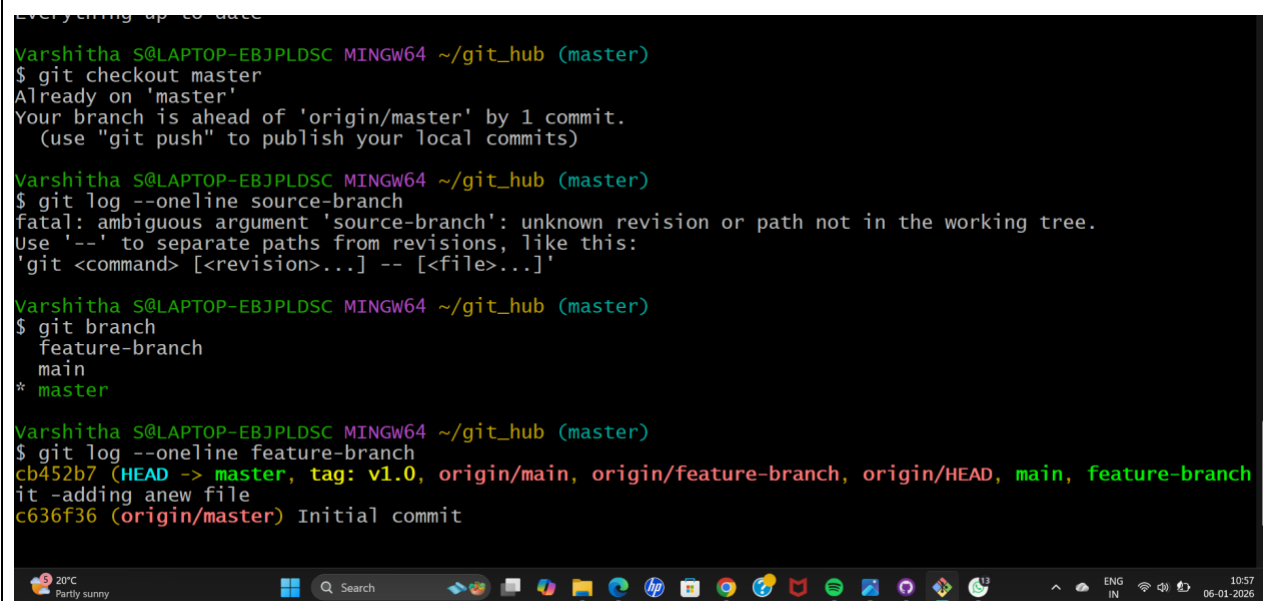
PROCEDURE:

1. The commit history was viewed using the git log --oneline command.
2. A new branch named feature1 was created using git checkout -b feature1.
3. The file lab1.txt was modified using Notepad.
4. The modified file was staged using the git add command.
5. The changes were committed using the git commit command.
6. The working branch was switched back to main using git checkout main.
7. The commit from the feature branch was applied to the main branch using the git cherrypick command.
8. The commit history was verified to confirm successful cherry-pick.

COMMANDS USED:

```
git log --oneline , git checkout -b feature1 notepad  
lab1.txt git add lab1.txt git commit -m "Commit for  
cherry-pick experiment" git checkout main git  
cherry-pick feature1  
git log --oneline
```

```
Everything up-to-date  
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)  
$ git checkout master  
Already on 'master'  
Your branch is ahead of 'origin/master' by 1 commit.  
(use "git push" to publish your local commits)  
  
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)  
$ git log --oneline source-branch  
fatal: ambiguous argument 'source-branch': unknown revision or path not in the working tree.  
Use '--' to separate paths from revisions, like this:  
'git <command> [<revision>...] -- [<file>...]'  
  
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)  
$ git branch  
  feature-branch  
  main  
* master  
  
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)  
$ git log --oneline feature-branch  
cb452b7 (HEAD -> master, tag: v1.0, origin/main, origin/feature-branch, origin/HEAD, main, feature-branch)  
it -adding anew file  
c636f36 (origin/master) Initial commit
```



EXPERIMENT 9: ANALYSING AND CHANGING GIT HISTORY

PROCEDURE:

1. The commit history of the repository was viewed using the git log --oneline command to display concise commit information.
2. The complete commit history with author name, date, and commit message was viewed using the git log command.
3. A specific commit was selected from the history and its detailed information was displayed using the git show command.
4. The changes made in the selected commit were analyzed using the diff output.

COMMANDS USED:

git log --oneline , git

log ,

git show <commit-id>

```
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git show abc123
fatal: ambiguous argument 'abc123': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git log -n 1 abc123
fatal: ambiguous argument 'abc123': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$
```

EXPERIMENT 10: ANALYSING AND CHANGING GIT

HISTORY

PROCEDURE:

1. The current status of the repository was checked using the git status command.
2. The file lab1.txt was modified using Notepad.
3. The repository status was checked again to confirm that the file was modified.
4. The modified file was restored to its last committed state using the git restore lab1.txt command.
5. The repository status was verified to ensure that the working tree was clean.

COMMANDS USED:

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git status

notepad lab1.txt

git status

```
varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git log --author="johnDoe"--since="2023-01-01"--until="2023-12-31"
```

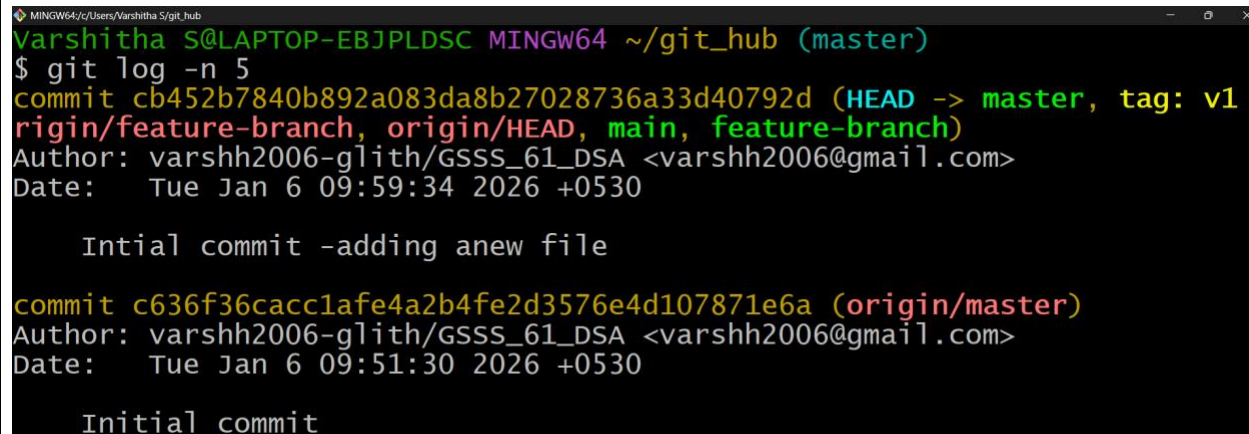
EXPERIMENT 11: ANALYSING AND CHANGING GIT HISTORY

PROCEDURE:

1. The current status of the local repository was checked using the git status command.
2. The committed changes were pushed to the remote GitHub repository using the git push origin main command.
3. After pushing, the status was verified to ensure the local and remote repositories were synchronized.
4. The updated files and commits were verified on the GitHub repository.

COMMAND USERS:

git status, git push origin



```
MINGW64/c/Users/Varshitha S/git_hub
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git log -n 5
commit cb452b7840b892a083da8b27028736a33d40792d (HEAD -> master, tag: v1
rigin/feature-branch, origin/HEAD, main, feature-branch)
Author: varshh2006-glith/GSSS_61_DSA <varshh2006@gmail.com>
Date: Tue Jan 6 09:59:34 2026 +0530

    Intial commit -adding anew file

commit c636f36cacc1afe4a2b4fe2d3576e4d107871e6a (origin/master)
Author: varshh2006-glith/GSSS_61_DSA <varshh2006@gmail.com>
Date: Tue Jan 6 09:51:30 2026 +0530

    Initial commit
```

EXPERIMENT 12: ANALYSING AND CHANGING GIT HISTORY

PROCEDURE:

1. The commit history was viewed using the git log --oneline command.
2. The commit that needed to be undone was identified.
3. The git revert command was used to reverse the changes made by the selected commit.
4. A new revert commit was created without deleting previous commits.
5. The repository status and commit history were verified.
6. The revert commit was pushed to the remote GitHub repository

COMMANDS USED:

git log --oneline, git revert <commit-id>, git status, git pus git push origin main


```
Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$ git revert abc123
fatal: bad revision 'abc123'

Varshitha S@LAPTOP-EBJPLDSC MINGW64 ~/git_hub (master)
$
```

git restore lab1.txt

git status